

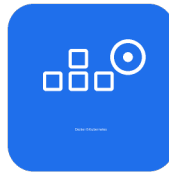


Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For



Application Integration with Docker and Kubernetes

TGS Ref No: TGS-2021010366

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 1.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	30 June 2026	First version — 2-day lesson plan (Docker + Kubernetes) aligned to the slide deck and the 19 hands-on labs; 9:00 AM-6:00 PM, 8 training hours/day	Tertiary Infotech Academy Pte Ltd

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Course Title	Application Integration with Docker and Kubernetes
Course Code	TGS-2021010366
Duration	2 Days (16 training hours)
Daily Schedule	9:00 AM – 6:00 PM (8 training hours/day, excluding lunch)
Lunch Break	1:00 PM – 2:00 PM (1 hour)
Breaks	Short tea breaks are scheduled within each day's training hours
Delivery Mode	Instructor-led, hands-on labs in KillerCoda browser terminals
Prerequisites	Basic command-line familiarity; no prior Docker or Kubernetes experience required
Tools	Docker, Docker Compose, kubectl, KillerCoda, a Docker Hub account

Course Overview

This 2-day hands-on course teaches participants to containerise, run and orchestrate real applications with Docker and Kubernetes. Throughout the course learners build and deploy a single sample web application — TaskBoard (a Flask task tracker backed by Redis and PostgreSQL) — progressing from a single container to a multi-service Docker Compose stack and finally a scalable Kubernetes deployment. Day 1 covers Docker: commands, building images with a Dockerfile, CMD vs ENTRYPOINT, volumes, networking, configuration, Docker Hub and Docker Compose. Day 2 covers Kubernetes: Pods, Namespaces, Deployments, rolling updates and rollbacks, Services, storage, and Jobs/CronJobs.

Learning Outcomes

By the end of this course, participants will be able to:

- Explain containerisation and the Docker architecture, and manage containers with the Docker CLI.
- Build custom images with a Dockerfile, applying layer caching and .dockerignore best practices.
- Distinguish CMD from ENTRYPOINT and package command-line tools correctly.
- Persist and share data using named volumes and bind mounts.
- Connect containers over a custom network using DNS, and publish services with port mapping.
- Configure containers at runtime with environment variables and publish images to Docker Hub.
- Define and run multi-service applications (web + cache + database) with Docker Compose.
- Explain Kubernetes architecture and deploy applications as Pods, Deployments and Services.

- Scale and self-heal workloads and perform zero-downtime rolling updates and rollbacks.
- Persist data with PersistentVolumes/Claims and run batch work with Jobs and CronJobs.

Day 1

Docker — Images, Storage, Networking & Compose

Morning Session · 9:00 AM – 1:00 PM

Time	Topic / Activity	Duration
9:00 – 9:15	Welcome, course objectives & overview (TGS-2021010366)	15 min
9:15 – 9:45	Topic 1: Containers, Docker architecture & core concepts	30 min
9:45 – 10:10	Lab 1: Docker Commands — Run & Manage Containers	25 min
10:10 – 10:35	Lab 2: Images & Inspecting Containers — pull, cp, inspect	25 min
10:35 – 10:50	<i>Tea Break</i>	15 min
10:50 – 11:10	Topic 2: Building images — Dockerfile, layers & build cache	20 min
11:10 – 11:45	Lab 3: Build the TaskBoard Image with a Dockerfile	35 min
11:45 – 12:10	Lab 4: Dockerfile Best Practices — .dockerignore & layer caching	25 min
12:10 – 12:45	Lab 5: CMD vs ENTRYPOINT — the taskboard-cli tool	35 min
12:45 – 1:00	Morning Q&A	15 min
Time	Topic / Activity	Duration
1:00 – 2:00	<i>Lunch Break</i>	60 min

Afternoon Session · 2:00 PM – 6:00 PM

Time	Topic / Activity	Duration
2:00 – 2:20	Topic 3: Docker storage — volumes & bind mounts	20 min
2:20 – 2:50	Lab 6: Docker Storage — Named Volumes & Bind Mounts	30 min
2:50 – 3:10	Topic 4: Docker networking — bridges, DNS & ports	20 min
3:10 – 3:40	Lab 7: Docker Networking — Custom Bridge, DNS & Port Mapping	30 min
3:40 – 3:55	<i>Tea Break</i>	15 min
3:55 – 4:15	Lab 8: Configuration with Environment Variables	20 min
4:15 – 4:35	Lab 9: Sharing Images — Push to Docker Hub	20 min
4:35 – 4:50	Topic 5: Docker Compose — multi-service apps	15 min
4:50 – 5:10	Lab 10: Docker Compose —	20 min

	Single Service	
5:10 – 5:30	Lab 11: Docker Compose — Multi-Service (Web + Redis)	20 min
5:30 – 5:55	Lab 12: Docker Compose — Full-Stack (Web + PostgreSQL + Redis)	25 min
5:55 – 6:00	Day 1 recap & wrap-up	5 min

Day 2

Kubernetes — Deploy, Scale, Update & Persist

Morning Session · 9:00 AM – 1:00 PM

Time	Topic / Activity	Duration
9:00 – 9:15	Day 1 recap & Day 2 objectives	15 min
9:15 – 9:55	Topic 6: Kubernetes architecture & kubectl	40 min
9:55 – 10:30	Lab 13: Kubernetes Pods — Imperative & Declarative	35 min
10:30 – 10:55	Lab 14: Kubernetes Namespaces — Environment Isolation	25 min
10:55 – 11:10	<i>Tea Break</i>	15 min
11:10 – 11:30	Topic 7: Deployments, ReplicaSets & self-healing	20 min
11:30 – 12:10	Lab 15: Deployments — Scaling & Self-Healing	40 min
12:10 – 12:45	Lab 16: Rolling Updates & Rollbacks	35 min
12:45 – 1:00	Morning Q&A	15 min
Time	Topic / Activity	Duration
1:00 – 2:00	<i>Lunch Break</i>	60 min

Afternoon Session · 2:00 PM – 6:00 PM

Time	Topic / Activity	Duration
2:00 – 2:20	Topic 8: Services & networking in Kubernetes	20 min
2:20 – 2:55	Lab 17: Services — ClusterIP & NodePort	35 min
2:55 – 3:10	Topic 9: Storage, Jobs & CronJobs	15 min
3:10 – 3:35	Lab 18: Kubernetes Storage — emptyDir, PV & PVC	25 min
3:35 – 4:00	Lab 19: Jobs & CronJobs — Batch and Scheduled Tasks	25 min
4:00 – 5:15	Final Assessment — written / MCQ on the LMS (lms.tertiaryinfotech.com)	75 min
5:15 – 5:30	<i>Tea Break</i>	15 min
5:30 – 6:00	Course review, learner feedback (TRAQOM) & closing	30 min

Labs Covered

All 19 hands-on labs are delivered in KillerCoda browser terminals and mirror the slide deck and Learner Guide exactly:

Lab	Topic	Title
1	Docker commands	Docker Commands — Run & Manage Containers
2	Docker commands	Images & Inspecting Containers — pull, cp, inspect
3	Docker Build / Dockerfile	Build the TaskBoard Image with a Dockerfile
4	Dockerfile best practices	Dockerfile Best Practices — .dockerignore & layer caching
5	CMD vs ENTRYPOINT	CMD vs ENTRYPOINT — the taskboard-cli tool
6	Volumes	Docker Storage — Named Volumes & Bind Mounts
7	Networking	Docker Networking — Custom Bridge, DNS & Port Mapping
8	Environment variables	Configuration with Environment Variables
9	Docker Hub	Sharing Images — Push to Docker Hub
10	Docker Compose	Docker Compose — Single Service
11	Docker Compose	Docker Compose — Multi-Service (Web + Redis)
12	Docker Compose	Docker Compose — Full-Stack (Web + PostgreSQL + Redis)
13	Pods	Kubernetes Pods — Imperative & Declarative
14	Namespaces	Kubernetes Namespaces — Environment Isolation
15	Deployments	Deployments — Scaling & Self-Healing
16	Rolling updates & rollbacks	Rolling Updates & Rollbacks
17	Services	Services — ClusterIP & NodePort
18	Storage	Kubernetes Storage — emptyDir, PV & PVC
19	Jobs & CronJobs	Jobs & CronJobs — Batch and Scheduled Tasks

Assessment

Participants are assessed through their hands-on lab work across both days and a final written / MCQ assessment on Day 2 (starting at 4:00 PM) delivered on the LMS. A minimum of 75% attendance across all sessions is required for SSG funding.